

Practice

Tags:

Time Limit: 1 s Memory Limit: 128 MB
 Submission: 2 AC: 1 Score: 99.94

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AI Code Tips

Description

How much does winning ACM depend on practice?

We assume that p , the probability that a given team will win a given contest, is related to n , the number of practice problems solved by the team prior to the contest. This relationship is modelled by the *logistic* formula

$$\text{Log}(p/(1-p)) = a + b n,$$

for some

a

and

b

. Your job is to find

a

and

b

such that the formula most accurately reflects a set of observed results.

Each observation consists of n and w . n is the number of practice problems solved by some team prior to a contest, and w is 1 if the team wins the contest, 0 if it does not.

Given a , b , and n the formula above may be used to compute p , the estimated probability that $w = 1$. The *likelihood* of a particular observation is p if $w = 1$ and $1-p$ if $w = 0$; The likelihood of a set of observations is the product of the likelihoods of the individual observations.

You are to compute the *maximum likelihood estimate* for a and b . That is, the values of a and b for which the likelihood of a given set of observations is maximized.

The input contains several test cases followed by a line containing 0. Each test case begins with $1 < k \leq 100$, the number of observations that follow. Each observation consists of integers $0 \leq n \leq 100$ and $0 \leq w \leq 1$. The input will contain at least two distinct values of n and of w . For each test case, output a single line containing a and b , rounded to four digits to the right of the decimal.

Samples

input	Copy
<pre>20 0 0 0 0 0 0 0 0 0 0 1 0 1 0 1 0 1 0 1 1 2 0 2 0 2 1 2 1 3 0 3 1 3 1 3 1 4 1 4 1 4 1 4 1 0</pre>	
output	Copy
<pre>-3.1748 1.5874</pre>	

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